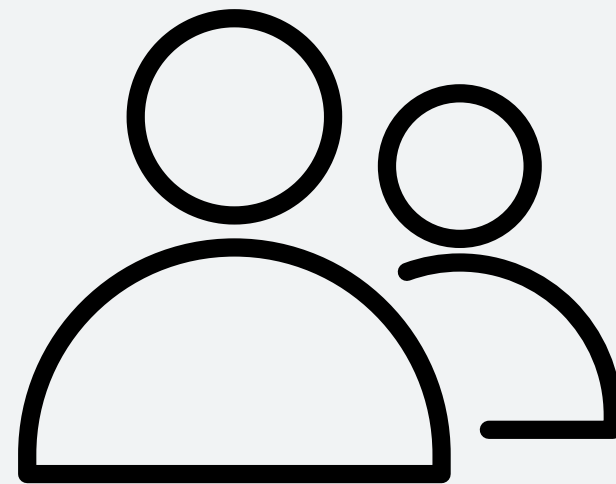


EMOTION PREDICTION FROM FACE AND VOICE WITH MACHINE LEARNING

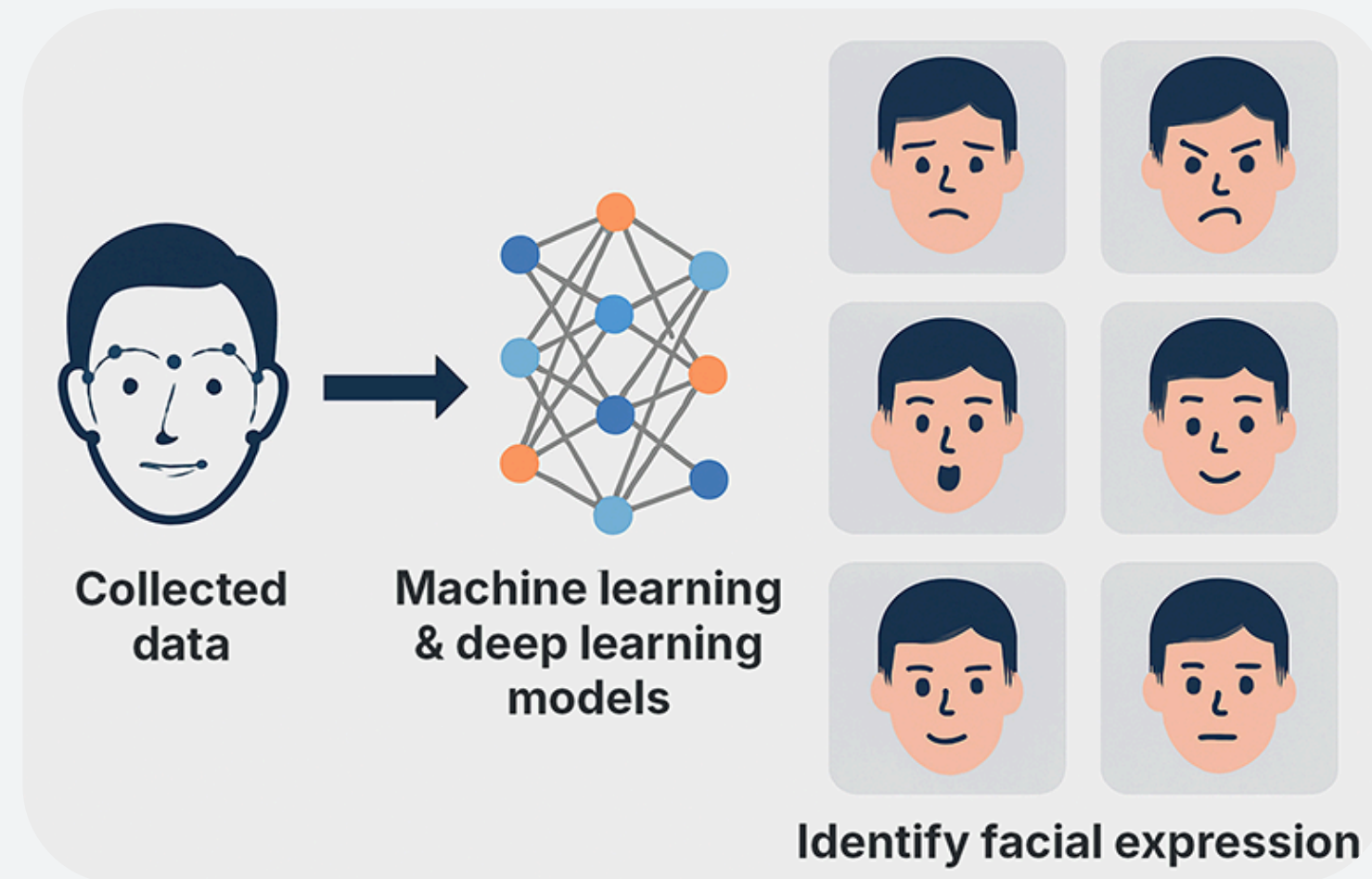
Introduction

Emotions play a central role in human communication. Not only what a person says, but also how they say it and the facial expression they carry while speaking directly affect the meaning of the conveyed message. While interpreting this information is a natural and intuitive process for humans in daily life, enabling computer systems to possess the same capability remains a challenging problem that researchers have been working on for decades.



Proje konusu

The objective of this study is to develop a multimodal, machine learning-based emotion recognition system that simultaneously analyzes human facial expressions and vocal dynamics. To overcome the inherent limitations of unimodal data sources, various architectures were designed, and a decision-level fusion strategy was implemented.



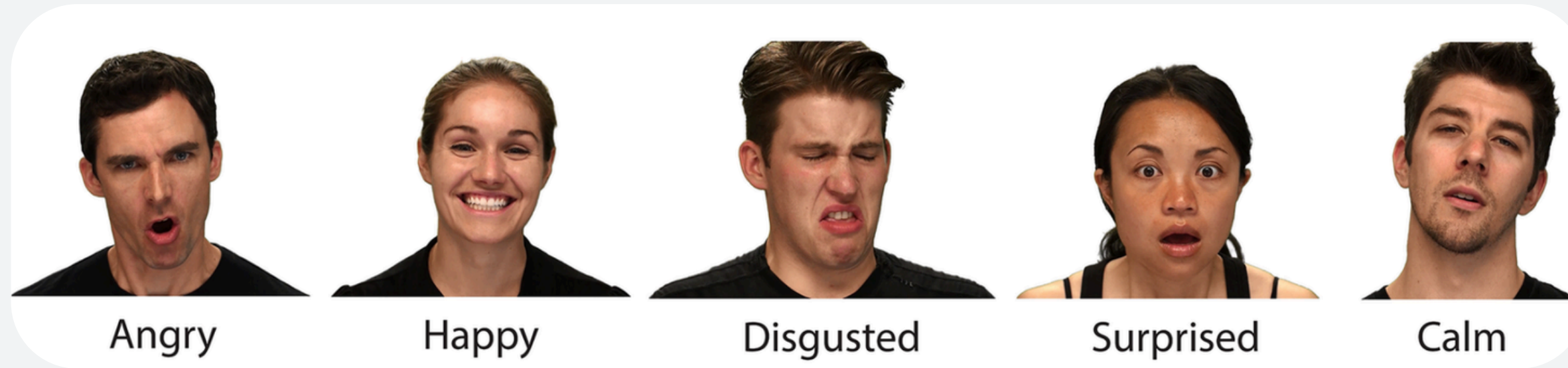
Başlıklar

- • Kullanılan Veri Seti
- • Ses Verisi Ön İşlemesi
- • Ses Modeli Mimarisi
- • Görüntü Verisi Ön İşlemesi
- • Görüntü Modeli Mimarisi
- • Birleştirilmiş Durum (Multi-Model)
- • Model Sonuçları
- • Karşılaştırma
- • Yorum

Dataset

RAVDESS Veri Seti

The dataset was generated by 24 professional actors, consisting of 12 females and 12 males. Each actor vocalized two predetermined sentences across eight distinct emotional states. These emotion classes include Neutral, Calm, Happy, Sad, Angry, Fearful, Disgust, and Surprised. Each recording was performed at two different intensity levels: normal and strong. Notably, the Neutral emotion class features only a single intensity level.



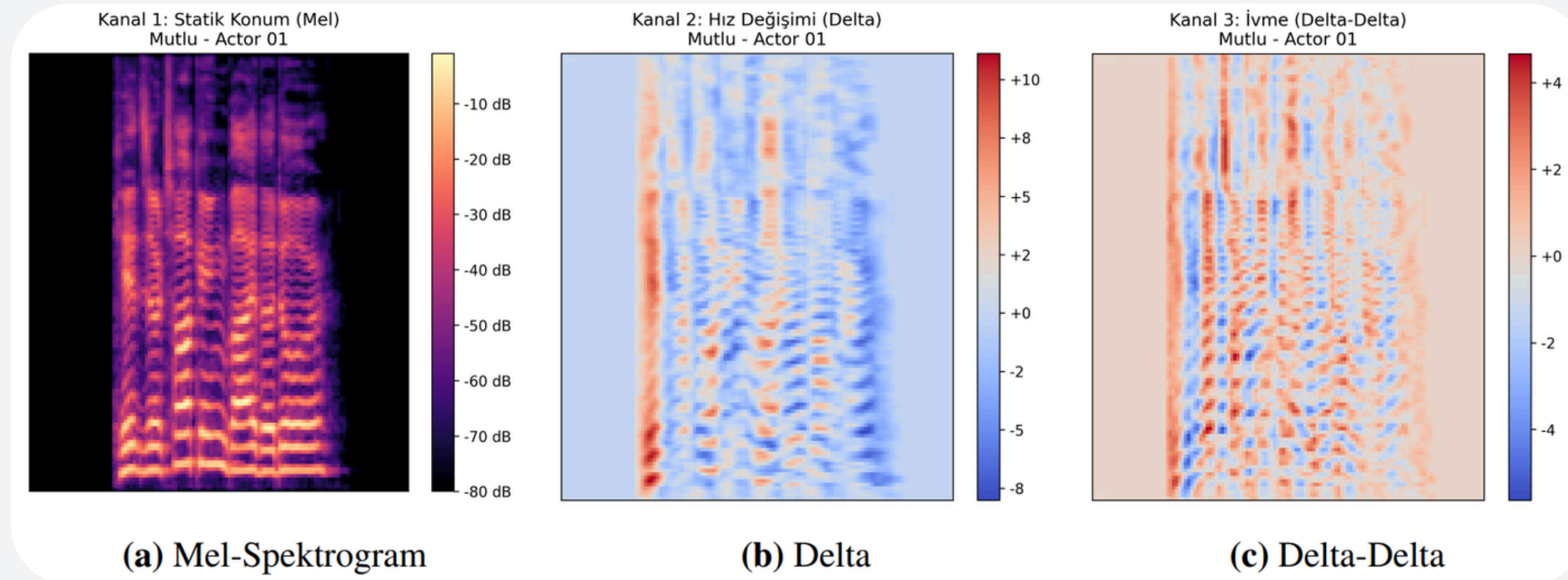
RAVDESS veri setinden örnek görüntüler.

Audio Data Preprocessing

In the first step, a Mel-Spectrogram was computed from the uploaded audio signal using 128 Mel bands. The resulting power spectrogram was then converted to the decibel scale. This transformation facilitates model learning by narrowing the dynamic range between low and high-energy components.

As the second channel, the delta (velocity) feature, which is the first-order derivative of the Mel-Spectrogram, was calculated.

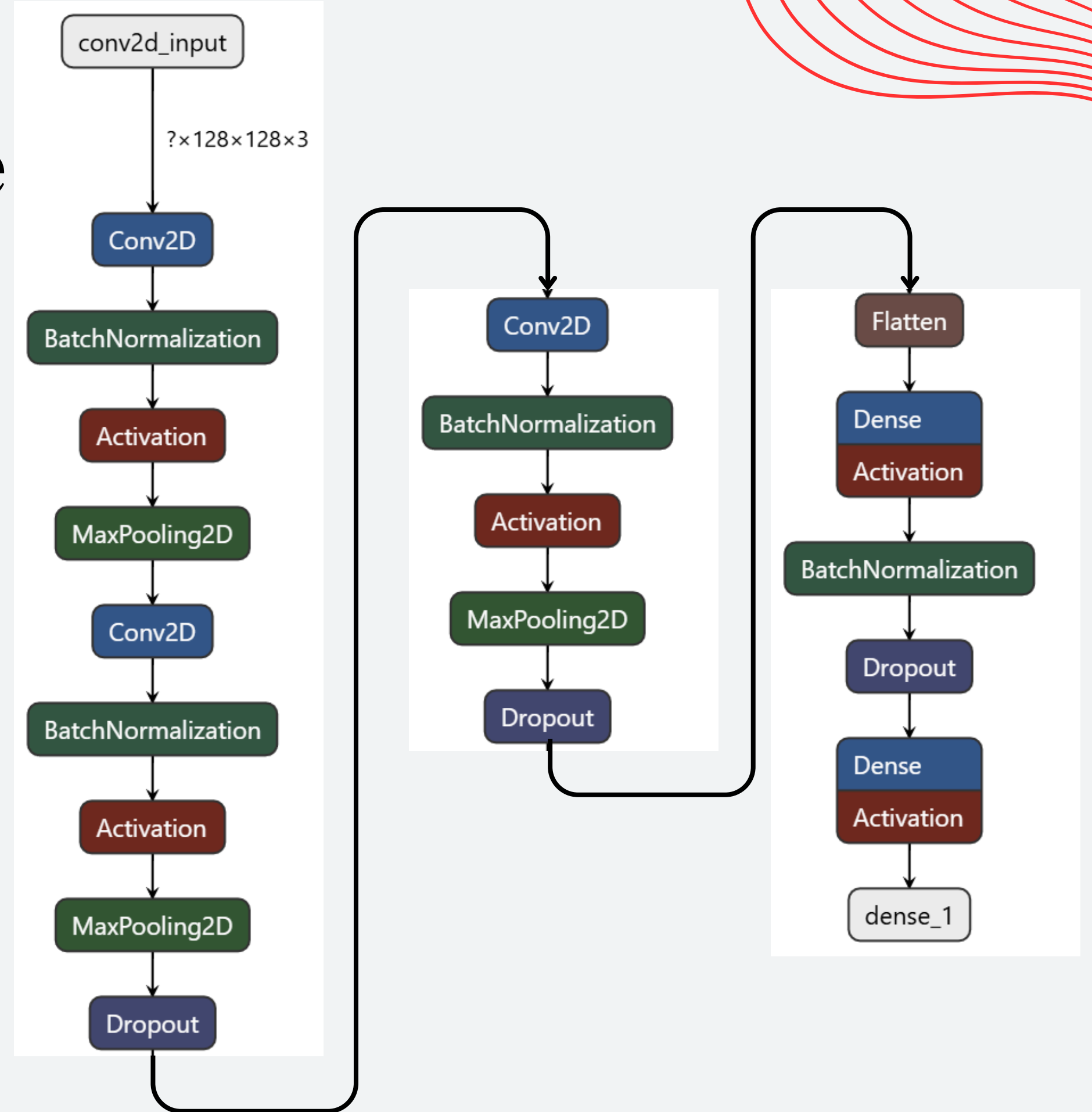
As the third channel, the delta-delta (acceleration) feature, representing the second-order derivative, was computed.



Mutlu duygu sınıfına ait bir ses kaydının üç kanallı görsel temsili.

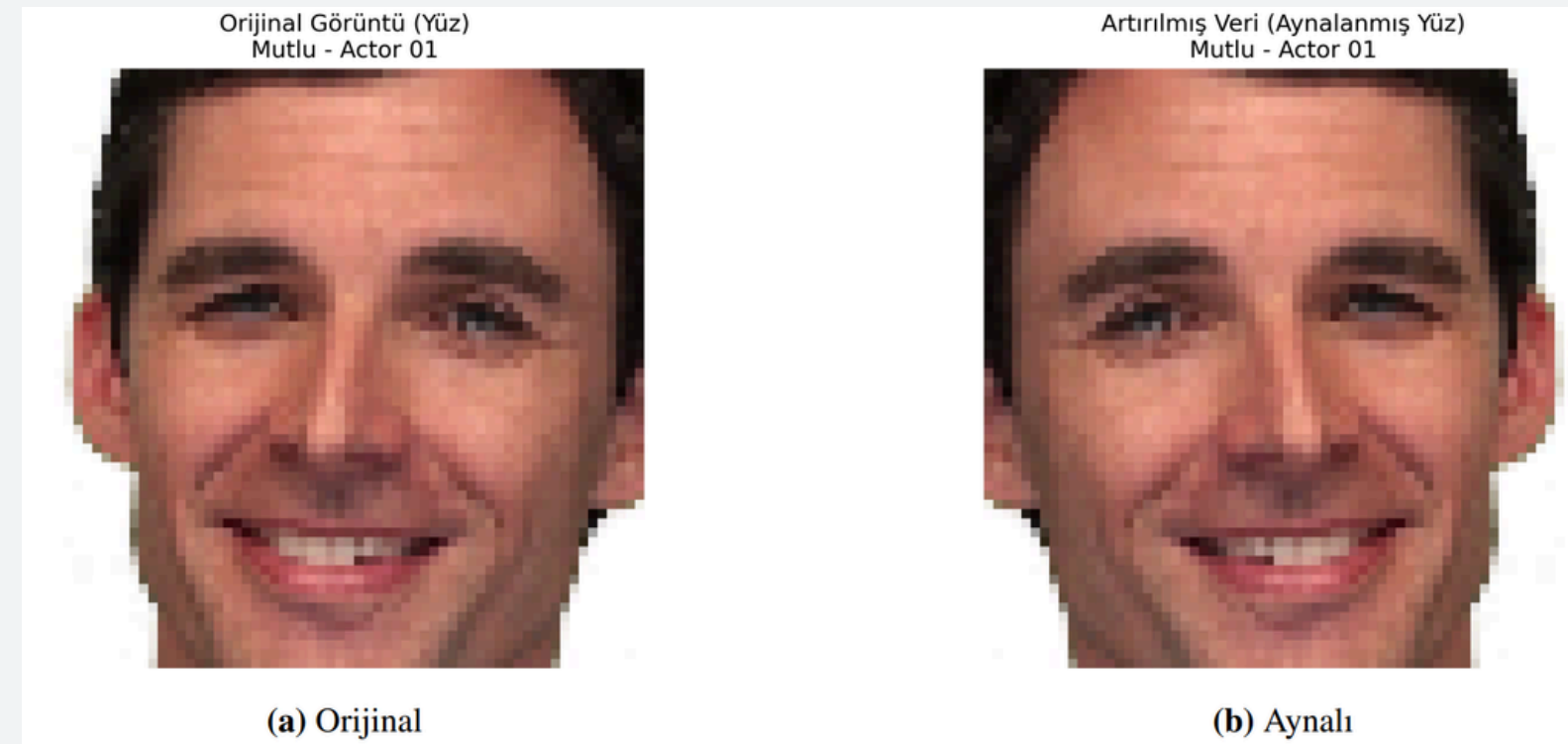
Audio Model Architecture

Ses duygu tanıma modeli, üç katmanlı bir Evrişimli Sinir Ağı (CNN) mimarisine sahiptir. Model, boyutundaki üç kanallı spektrogram girdisini alarak sekiz duygu sınıfı için olasılık dağılımı üretmektedir.



Visual Data Preprocessing

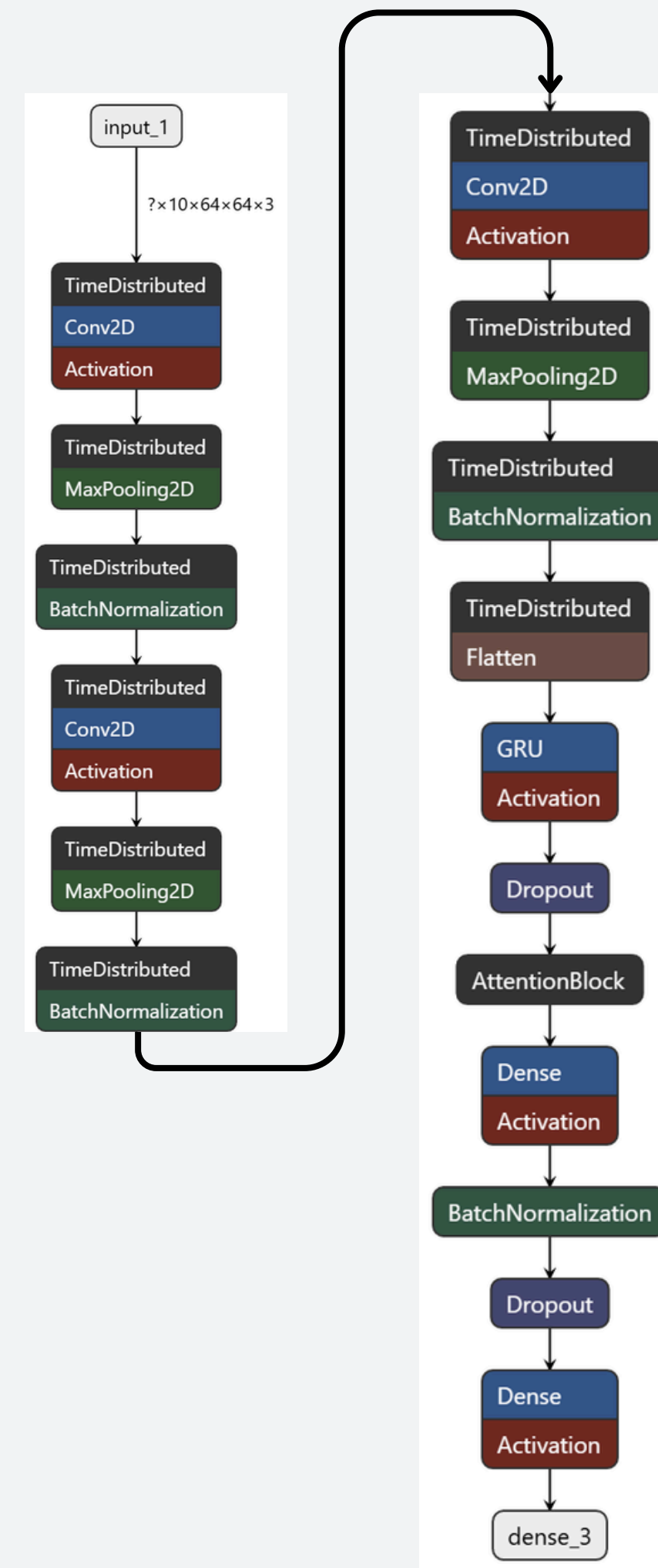
To perform emotion recognition from video files, a fixed number of frames must first be extracted from each video. In this study, 10 frames were sampled at equal intervals from each video. Face detection was performed on each selected frame using the Haar Cascade algorithm, and the resulting faces were resized to 64x64 pixels. To expand the training dataset, data augmentation was applied using horizontal flipping. A horizontally flipped copy was generated for every original facial sequence, effectively doubling the volume of the training data.



Mutlu duygu sınıfına ait yüz görüntüsünün orijinal ve aynalı versiyonu

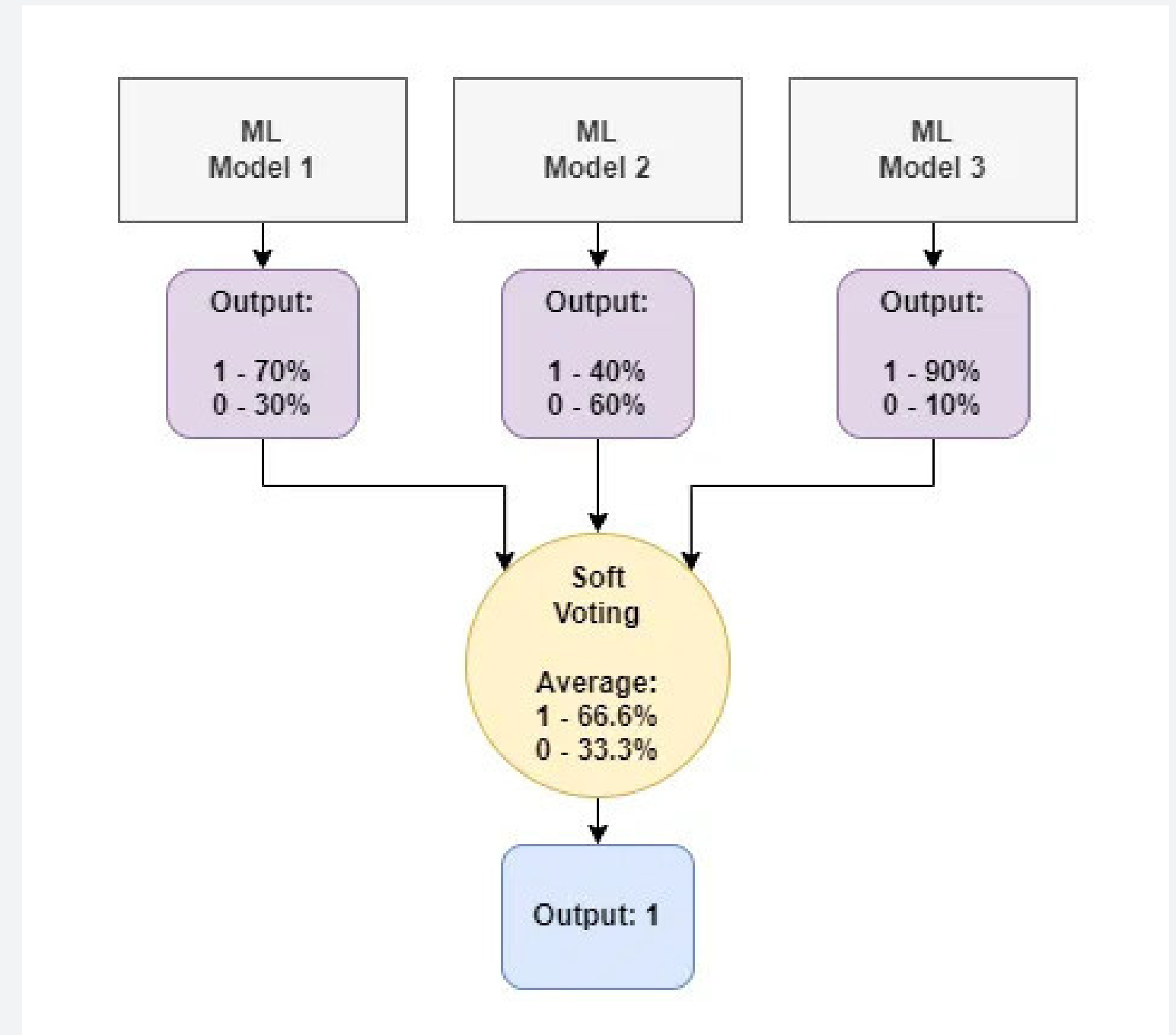
Image Model Architecture

Görüntü duygu tanıma modeli; Time Distributed CNN, GRU ve Attention katmanlarını birleştiren hibrit bir mimari üzerine inşa edilmiştir. Bu mimari, yüz ifadelerinin hem uzamsal özelliklerini hem de zaman içindeki değişimlerini birlikte modellemektedir.



Combined Multimodal System

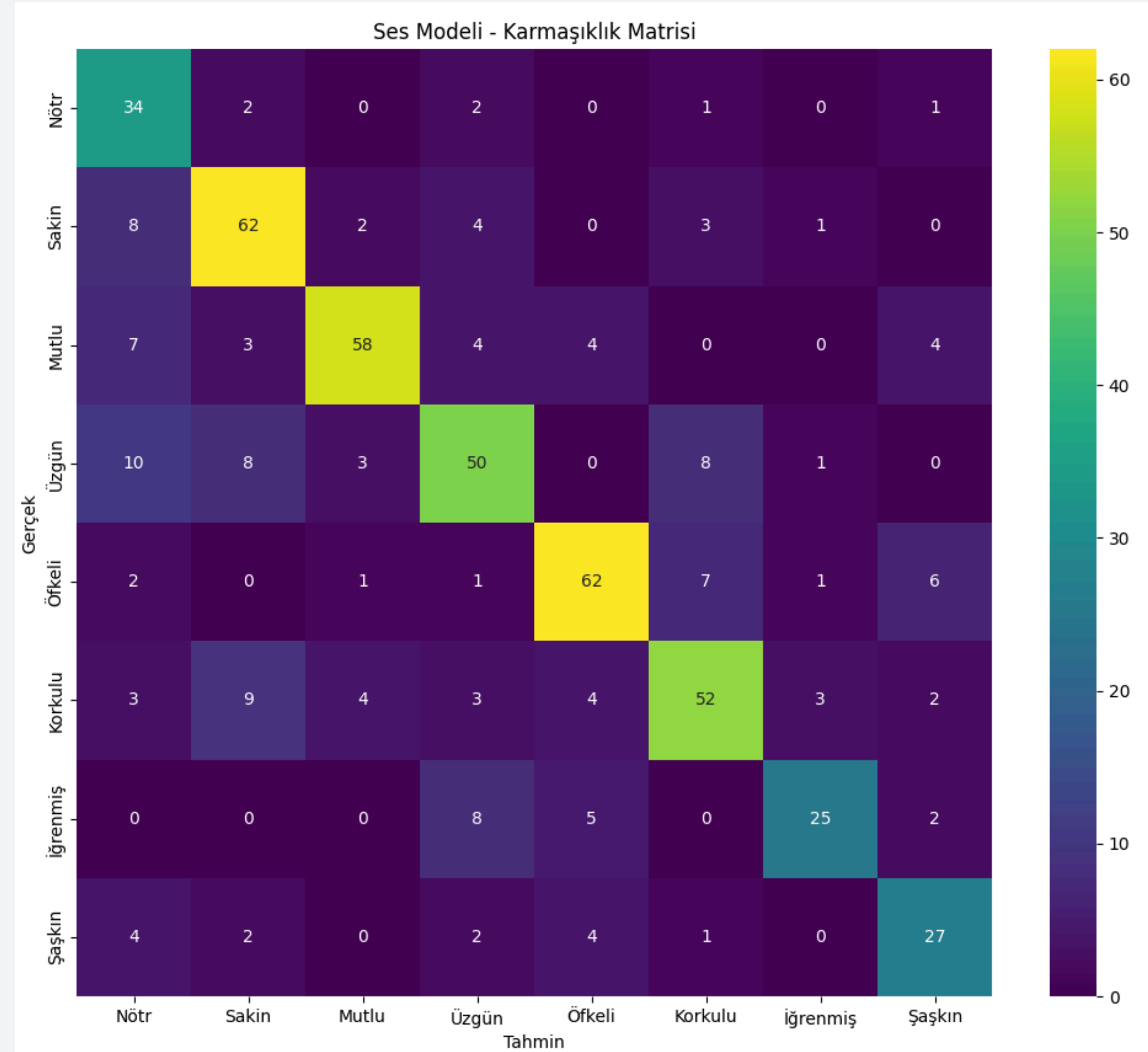
A late fusion approach was adopted to combine the outputs of the audio and image models. In this method, both models are trained independently, and their outputs are fused during the testing phase. The Softmax output layers of both models generate probability values for each of the eight emotion classes. The combined prediction is then calculated by taking the weighted average of these probability vectors via soft voting.



Soft Voting görsel temsili.

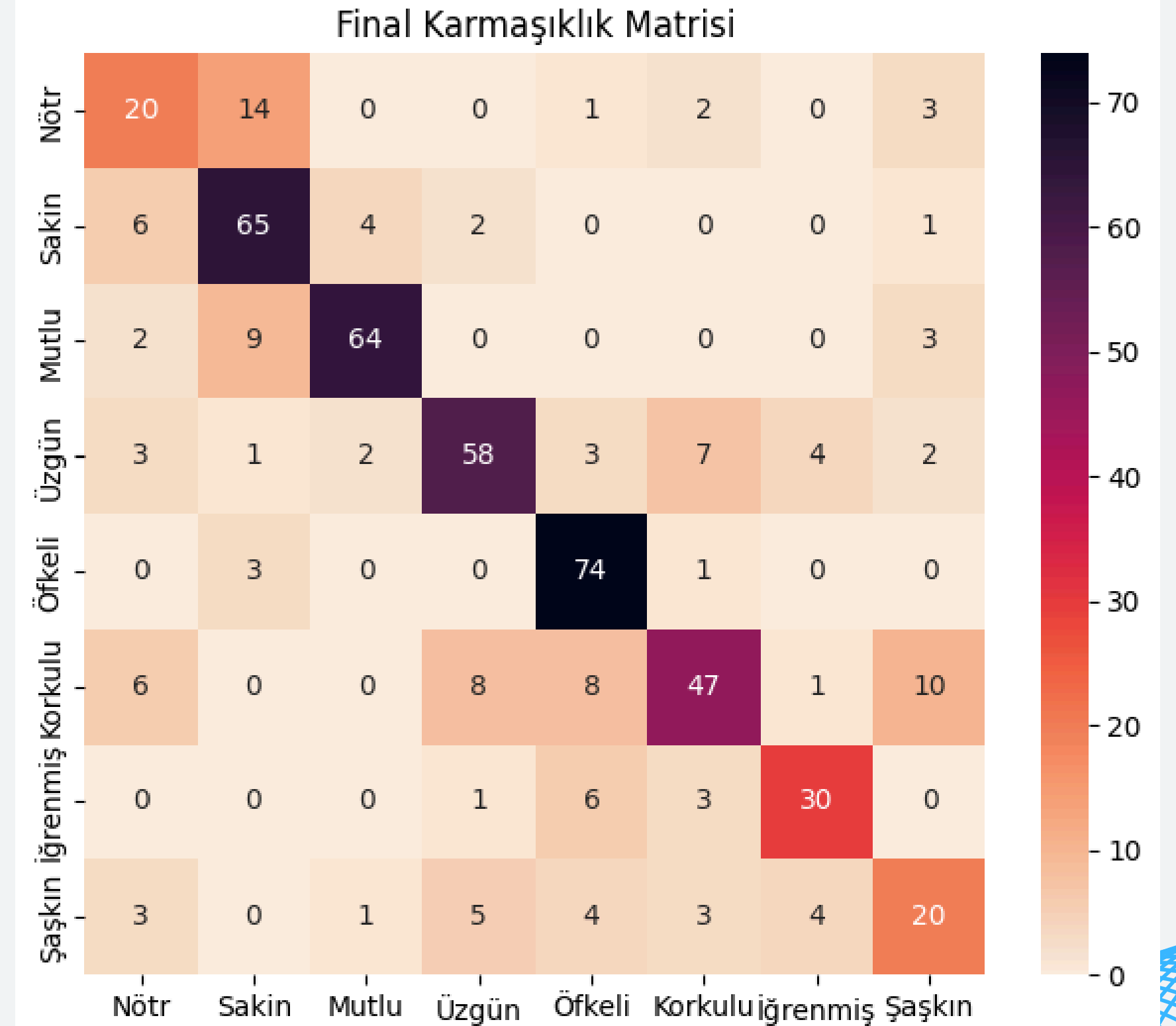
Sonuçlar - *Ses Modeli*

	precision	recall	f1-score	support
Nötr	0.50	0.85	0.63	40
Sakin	0.72	0.78	0.75	80
Mutlu	0.85	0.72	0.78	80
Üzgün	0.68	0.62	0.65	80
Öfkeli	0.78	0.78	0.78	80
Korkulu	0.72	0.65	0.68	80
İğrenmiş	0.81	0.62	0.70	40
Şaşkın	0.64	0.68	0.66	40
accuracy			0.71	520
macro avg	0.71	0.71	0.70	520
weighted avg	0.73	0.71	0.71	520



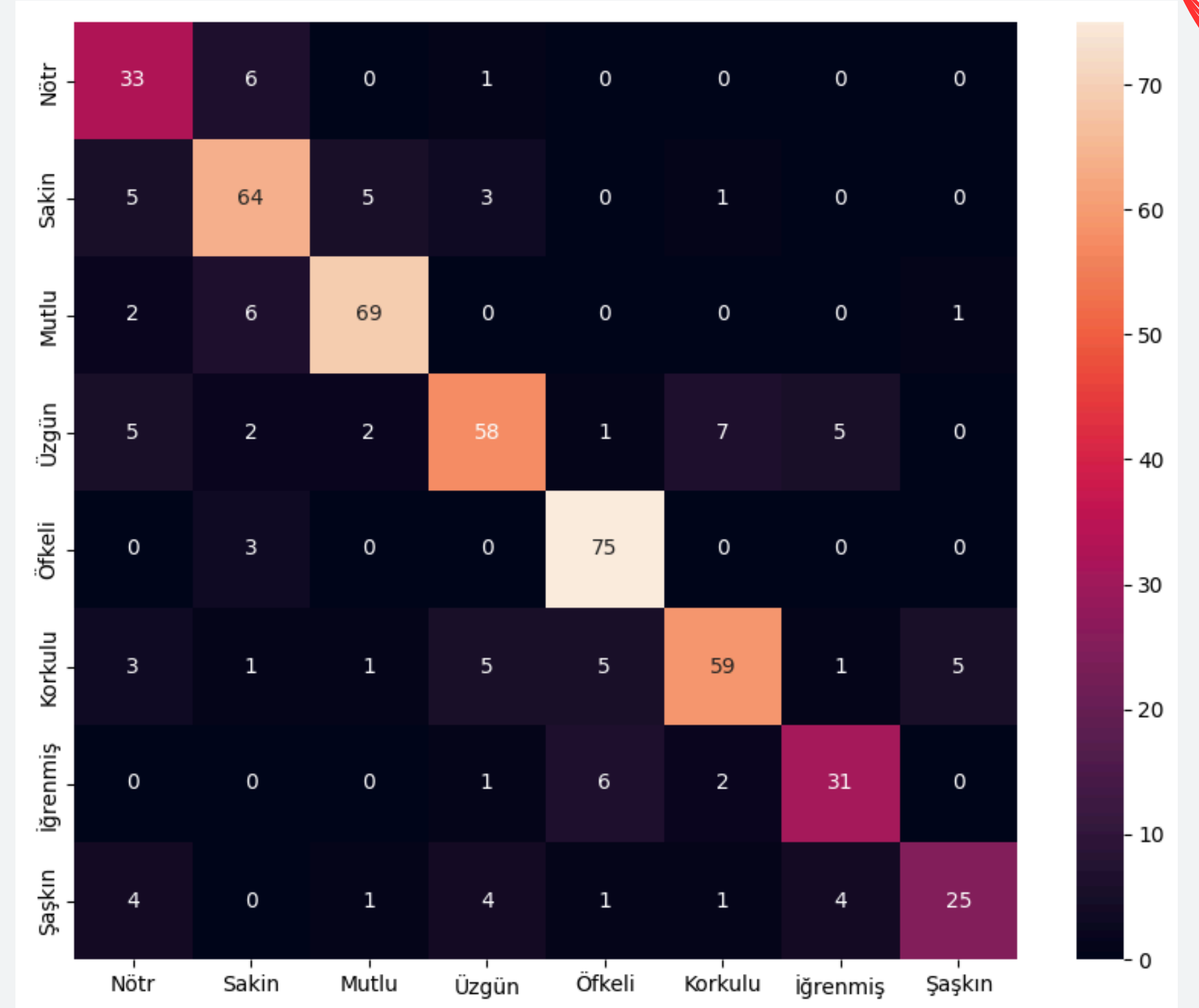
Sonuçlar - Görüntü Modeli

	precision	recall	f1-score	support
Nötr	0.50	0.50	0.50	40
Sakin	0.71	0.83	0.76	78
Mutlu	0.90	0.82	0.86	78
Üzgün	0.78	0.72	0.75	80
Öfkeli	0.77	0.95	0.85	78
Korkulu	0.75	0.59	0.66	80
İğrenmiş	0.77	0.75	0.76	40
Şaşkın	0.51	0.50	0.51	40
accuracy			0.74	514
macro avg	0.71	0.71	0.71	514
weighted avg	0.74	0.74	0.73	514



Sonuçlar - Birleştirilmiş Durum (Multi-Model)

	precision	recall	f1-score	support
Nötr	0.63	0.82	0.72	40
Sakin	0.78	0.82	0.80	78
Mutlu	0.88	0.88	0.88	78
Üzgün	0.81	0.72	0.76	80
Öfkeli	0.85	0.96	0.90	78
Korkulu	0.84	0.74	0.79	80
İğrenmiş	0.76	0.78	0.77	40
Şaşkın	0.81	0.62	0.70	40
accuracy			0.81	514
macro avg	0.80	0.79	0.79	514
weighted avg	0.81	0.81	0.80	514



Comparison

Model	Genel Doğruluk (%)	Ağırlıklı F1	Makro F1
Ses Modeli	71	0.71	0.70
Görüntü Modeli	74	0.73	0.71
Birleşik Model	81	0.80	0.79

Duygu	Ses F1	Görüntü F1	Birleşik F1
Nötr	0.63	0.50	0.72
Sakin	0.75	0.76	0.80
Mutlu	0.78	0.86	0.88
Üzgün	0.65	0.75	0.76
Öfkeli	0.78	0.85	0.90
Korkulu	0.65	0.66	0.79
İğrenmiş	0.70	0.76	0.77
Şaşkın	0.66	0.51	0.70

Aktör Bazlı Ayırmanın Önemi

	precision	recall	f1-score	support
Nötr	0.83	0.92	0.88	38
Sakin	0.91	0.96	0.94	75
Mutlu	0.88	0.89	0.89	75
Üzgün	0.78	0.81	0.80	75
Öfkeli	0.91	0.93	0.92	75
Korkulu	0.95	0.69	0.80	75
İğrenmiş	0.77	0.95	0.85	39
Şaşkın	0.89	0.82	0.85	39
accuracy			0.87	491
macro avg	0.87	0.87	0.86	491
weighted avg	0.87	0.87	0.87	491

Ses Modeli, Duygu bazlı ayırma sonuçları

	precision	recall	f1-score	support
Nötr	0.81	0.92	0.86	38
Sakin	0.88	0.86	0.87	73
Mutlu	0.95	0.97	0.96	74
Üzgün	0.93	0.88	0.90	75
Öfkeli	0.90	0.96	0.93	73
Korkulu	0.93	0.88	0.90	75
İğrenmiş	0.97	0.87	0.92	38
Şaşkın	0.87	0.89	0.88	38
accuracy			0.91	484
macro avg	0.90	0.90	0.90	484
weighted avg	0.91	0.91	0.91	484

Görüntü Modeli, Duygu bazlı ayırma sonuçları

Yorum

- Geliştirilen çok modlu duygu tanıma sistemi, ses modeli ile %71, görüntü modeli ile %74 ve birleşik model ile %81 genel doğruluk elde etmiştir. Birleşik modelin tek başına çalışan modellere göre belirgin bir performans artışı sağlaması, ses ve görüntü modellerinin birbirini tamamlayıcı bilgi taşıdığını göstermektedir.
- Eğitim ve test setleri aktör bazlı olarak ayrılmış ve aynı aktörün verileri her iki sette yer almamıştır. Bu yöntem, rastgele bölmeye kıyasla daha güvenilir sonuçlar üretmektedir. Bunun sebebi modelin kişileri ezberlemesini engellemektir.



Dinlediğiniz için Teşekkürler